Shaping Latent World Models via Latent Reasoning for Embodied Planning and Generalization

1 Research Problem

We study action-conditioned world modeling for embodied agents: given an observation image o_t and an action a_t, predict the next world state. Our core hypothesis is that a *latent token block* can serve as a unified *state carrier* that fuses vision, language context, action-conditioning, and future prediction. Concretely, we want a multimodal LLM/VLM to generate a structured latent block $\hat{z}_{t+1}$ that represents the next state. We then ground $\hat{z}_{t+1}$ to real transitions using simulator trajectories and image-level supervision (Fig. 1).

2 Why This Problem Is Important

Embodied agents require representations that are simultaneously (i) grounded in visual observations, (ii) conditioned on actions, and (iii) predictive of future consequences. If the internal state is not aligned with real environment transitions, multi-step rollouts quickly drift and cannot support downstream decision making. A latent "state carrier" trained from large-scale transition data provides a principled interface for integrating perception, action, and future prediction. This also yields a testable artifact: rollout quality can be directly evaluated by comparing predicted next observations against ground truth.

3 Existing Work and Limitations

Latent world models. Recent work learns world models for control and robotics, including predictive world models without reconstruction and compositional robot imagination [2, 14] as well as large-scale interactive environment models [1]. However, it remains unclear how to train a *structured latent state* that is (i) emitted reliably by an LLM/VLM and (ii) explicitly grounded to action-conditioned transitions.

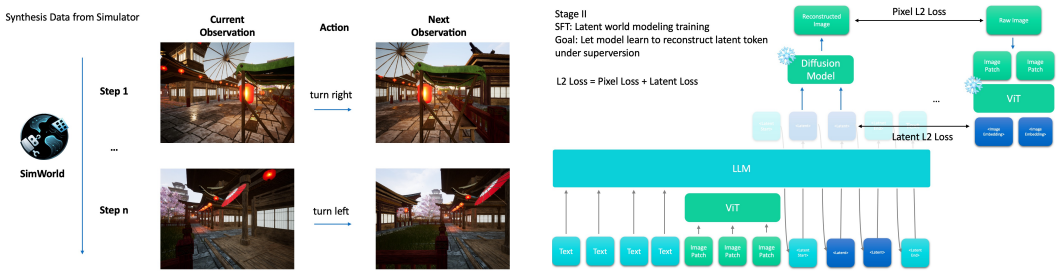
Latent reasoning. Hidden/latent reasoning methods study how to perform multi-step computation in latent spaces rather than explicit token-level rationales [5, 3, 9, 12, 13, 10, 7]. In this proposal, we adopt the key idea of *latent computation* but focus it on *world-state prediction*: the latent block is trained to represent z_{t+1} under action-conditioned transitions.

Benchmarks and simulators. BEHAVIOR-1K and recent evaluation suites emphasize long-horizon embodied activities and systematic benchmarking of embodied agents/world models [8, 4, 6]. To support scalable interaction data collection for world-model learning, recent simulators such as SimWorld provide realistic, open-ended environments and agent interfaces for LLM/VLM agents [11].

4 Proposed Solution

Latent block as next-state. We model action-conditioned transitions by predicting a structured latent token block $\hat{z}_{t+1} =$ `<latent_start>` $\hat{z}^{(1)} \dots \hat{z}^{(K)}$ `<latent_end>` and treat it as the next world state. The latent block is the *state carrier*: it integrates multimodal context and provides an interface for multi-step world modeling. Importantly, we do not treat the latent block as an explanation; it is trained to be a predictive state variable anchored to environment transitions.

Architecture. A ViT encodes o_t into patch tokens; the LLM consumes text context + vision tokens and outputs a latent block $\hat{z}_{t+1}$. A diffusion decoder maps $\hat{z}_{t+1} \mapsto \hat{o}_{t+1}$ to provide image-space supervision and qualitative diagnostics.



(a) SimWorld data collection (tuples o_t, a_t, o_{t+1}).

(b) Stage II training: decode $\hat{z}_{t+1}$ to $\hat{o}_{t+1}$ and apply pixel/latent losses.

Figure 1: Overview of data collection in SimWorld (left) and Stage II world-model supervision (right).

Stage I: latent-block formatting. We SFT the model to reliably generate `<latent_start>...<latent_end>` blocks using standard cross-entropy over target sequences. Concretely, the training target includes both text tokens and a fixed-length latent block, so the model learns (a) where to start/end the latent channel and (b) how to keep latent length bounded.

Stage II: world-model supervision (Fig. 1b). Using SimWorld-collected tuples (o_t, a_t, o_{t+1}) [11], we train the latent block to represent the next observation. We feed (o_t, a_t) to the model, obtain $\hat{z}_{t+1}$, decode it to an image $\hat{o}_{t+1}$ with a diffusion decoder, and supervise with $\mathcal{L} = \|\hat{o}_{t+1} - o_{t+1}\|_2^2 + \lambda\|e(\hat{o}_{t+1}) - e(o_{t+1})\|_2^2$, where $e(\cdot)$ is the ViT embedding (we will study freezing vs. finetuning in ablations). This combines low-level geometric/appearance matching with high-level semantic alignment. In practice, we will sweep λ and ablate whether the diffusion decoder and/or ViT is frozen to understand where the supervision signal is most effective.

Evaluation plan. After training, the model outputs $\hat{z}_{t+1}$ for a given (o_t, a_t) and can be iterated to produce multi-step predicted latent trajectories. We will evaluate both one-step prediction and multi-step rollout drift on multiple settings: (i) SimWorld trajectories [11] as the primary transition-supervision and in-domain evaluation source, and (ii) BEHAVIOR-1K / EmbodiedEval-style protocols [8, 4] (plus EWMBench for world-model quality stress tests [6]). Metrics will include pixel error, ViT-embedding error, and multi-step consistency under fixed action sequences.

5 Novelty and Significance

- **Latent space as a state carrier.** We treat a latent token block as the explicit representation of world state, unifying visual evidence, action-conditioning, and future prediction in a single interface.
- **Latent channel stabilization.** A dedicated Stage I ensures the model can reliably emit well-formed latent blocks before any world-model grounding.
- **Grounding via transition supervision.** Stage II anchors the latent state to real dynamics using (o_t, a_t, o_{t+1}) supervision from SimWorld [11], with pixel and embedding losses.
- **Clear deliverables.** We will report transition fidelity and multi-step rollout consistency on BEHAVIOR-1K / EmbodiedEval-style evaluations [8, 4], including one-step reconstruction error and multi-step drift.

6 Tentative Timeline

1. **W1:** Set up SimWorld data pipeline and evaluation on BEHAVIOR-1K / EmbodiedEval.
2. **W2–W3:** Stage I SFT for latent-block format; Stage II supervision with diffusion decoding.
3. **W4:** Multi-step rollout evaluation; ablations (freeze vs. finetune; loss weights).
4. **W5:** Generalization and robustness tests (new scenes, action distributions).
5. **W6:** Final experiments, figures, and write-up.

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